A Hybrid Machine Learning Approach for Fraud Detection Using Autoencoders and XGBoost

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Abstract

Financial fraud detection remains a major challenge in modern banking systems [1] due to the extreme class imbalance present in transactional data and the increasingly sophisticated strategies used by fraudsters. Traditional supervised learning models often struggle to detect zero-day or previously unseen fraud patterns, while purely unsupervised approaches may generate a high number of false positives, leading to customer dissatisfaction and operational inefficiency.Recent studies have shown that combining anomaly detection with optimized machine learning classifiers significantly improves fraud detection performance [2].

To address these limitations, this project proposes a multi-tiered hybrid framework designed to improve both detection performance and system reliability. The framework is evaluated using the AIML Financial Dataset and incorporates three complementary approaches: a Logistic Regression model as a baseline classifier, a deep Autoencoder for unsupervised anomaly detection, and a hybrid model based on Gradient Boosting (XGBoost) leveraging learned feature representations.

The results demonstrate that combining deep representation learning with ensemble-based classification improves detection accuracy while reducing false positives, making the system more effective for real-world financial fraud detection scenarios.

Keywords

Fraud Detection • Financial Technology (FinTech) • Anomaly Detection • Deep Autoencoders • XGBoost • Class Imbalance • Hybrid Machine Learning

1 Introduction

Financial fraud has become one of the most critical challenges in modern banking and financial systems due to the rapid growth of digital transactions. With the increasing adoption of online payment systems, fraudsters continuously develop more sophisticated and adaptive techniques, making traditional rule-based detection systems ineffective.Financial fraud detection has become one of the most important applications of machine learning due to the increasing volume of digital transactions.

A major difficulty in fraud detection is the highly imbalanced nature of transactional datasets, where legitimate transactions significantly outnumber fraudulent ones. This imbalance often leads machine learning models to become biased toward the majority class, resulting in poor detection of fraudulent activities. Additionally, many fraudulent transactions are rare and evolve over time, making it difficult for purely supervised learning models to generalize well to unseen or "zero-day" fraud patterns.

To address these limitations, machine learning and deep learning techniques have been widely adopted in recent years. Supervised models such as Logistic Regression provide strong baseline performance and interpretability but often fail to capture complex nonlinear patterns in data. On the other hand, unsupervised deep learning methods such as Autoencoders are effective in learning compressed feature representations and detecting anomalies through reconstruction error, making them suitable for identifying previously unseen fraud patterns.

However, each approach has its own limitations when used independently. Logistic Regression lacks expressive power for complex relationships, while Autoencoders may produce higher false positive rates when used alone for classification tasks. To overcome these challenges, ensemble and hybrid approaches have gained significant attention. In particular, Gradient Boosting methods such as XGBoost have demonstrated strong performance in handling structured data, class imbalance, and nonlinear feature interactions.Several machine learning algorithms, including Random Forest, XGBoost, and neural networks, have been successfully applied to banking fraud detection [3].

In this project, we propose a hybrid fraud detection framework that integrates Logistic Regression, Autoencoders, and XGBoost to leverage the strengths of both supervised and unsupervised learning techniques. The Autoencoder is used for unsupervised feature learning and anomaly detection, while XGBoost is employed as a powerful classifier that operates on enriched feature representations. Logistic Regression is used as a baseline model for comparison and interpretability.

The proposed system aims to improve detection accuracy, reduce false positives, and enhance robustness against evolving fraud patterns. The effectiveness of the framework is evaluated using the PaySim Financial Transaction Dataset, and the results demonstrate the advantages of combining deep learning-based feature extraction with gradient boosting classifiers for fraud detection tasks.

1.1 Research Questions

This study investigates whether combining unsupervised anomaly detection with supervised ensemble learning can improve fraud detection performance in highly imbalanced financial transaction datasets. Specifically, the following research questions are addressed:

RQ1: How effectively can a deep Autoencoder detect anomalous financial transactions compared to a traditional supervised baseline model?

RQ2: Does integrating Autoencoder-generated anomaly scores with supervised classifiers improve fraud detection performance?

RQ3: Can a hybrid Autoencoder-XGBoost framework achieve a better balance between precision, recall, and false positive reduction than standalone machine learning models?

2 Data Collection Exploratory Data Analysis (EDA)

2.1 PaySim Simulation Environment

The dataset used in this work is the PaySim financial transaction dataset, which simulates real-world mobile money transactions and contains both legitimate and fraudulent activities [4]. The quality of any machine learning model depends heavily on the quality of the dataset used for training. In financial applications, obtaining real transaction data is difficult because of privacy regulations, security concerns, and confidentiality requirements.

To overcome these challenges, researchers often use synthetic datasets that mimic real-world financial behavior while protecting sensitive customer information.

The PaySim dataset is one such benchmark dataset. It was created using agent-based simulation techniques that model realistic financial transactions observed in mobile money services. To conduct our experiments safely without compromising sensitive banking data, we utilize the PaySim mobile money simulation dataset. PaySim uses agent-based modeling to generate synthetic financial records based on real aggregate logs from an operational African cellular payment service.

Dataset Size: 6,362,620 observations. Feature Scope: 11 distinct multi-modal columns detailing individual cash flow attributes.

Transaction type
Transaction amount
Sender balance before transaction
Sender balance after transaction
Receiver balance before transaction
Receiver balance after transaction
Fraud label

The dataset provides an excellent environment for developing and evaluating fraud detection algorithms.

2.2 Exploratory Insights Behavioral Patterns

Exploratory Data Analysis (EDA) was performed to understand the distribution of transactions, identify missing values, and detect potential outliers before model training [5]..

The primary objectives of EDA include:

Identifying missing values
Detecting outliers
Understanding feature distributions
Examining class imbalance
Discovering hidden patterns

During exploratory analysis, we tracked the distribution of fraudulent actions across different transaction types.

The analysis revealed an important operational detail: Fraudulent activity is exclusively concentrated within two transaction types: TRANSFER and CASH-OUT.

This structural pattern reflects real-world financial fraud behavior:

First, an account is compromised or taken over, and funds are moved out via a TRANSFER to an intermediary shell account.

Second, an accomplice rapidly cashes out the stolen funds through a physical CASH-OUT vendor before internal security flags them.

This insight allowed us to focus our feature engineering and modeling directly on these two high-risk operational modes.

3 Data Preprocessing and Feature Engineering

Raw simulation logs cannot be fed directly into deep neural networks or gradient boosting models without preparation. We developed a structured preprocessing pipeline to clean the data and extract meaningful feature representations.

Raw transactional data often contains inconsistencies, noise, and redundant information. Machine learning algorithms require clean and properly formatted data to achieve optimal performance.

Data preprocessing generally involves:

Data cleaning
Handling missing values
Feature scaling
Encoding categorical variables
Removing duplicate records

3.1 Feature Engineering

Feature engineering was performed to derive meaningful attributes from raw transaction data and improve the model's ability to identify fraudulent behavior.

To help our models capture sudden, abnormal variations in account balances rather than looking at transaction amounts alone, we engineered two continuous balance-tracking features:Origin Account Balance Discrepancy

(Δ_{orig}):

$$\Delta_{\text{orig}} = \text{oldbalanceOrg} - \text{newbalanceOrig}$$

Rationale: For genuine transfers, this difference typically matches the amount column exactly. For fraudulent transactions, this value often highlights internal clearing discrepancies or unauthorized overdraft overrides.Destination Account Balance Discrepancy

(Δ_{dest}):

$$\Delta_{\text{dest}} = \text{newbalanceDest} - \text{oldbalanceDest}$$

Rationale: A common fraud pattern involves transferring huge sums into an account that immediately clears the funds out to zero. Tracking Δ_{dest} relative to the transaction amount exposes these illicit matching anomalies.

3.1.1 Ratio-Based Features. A. Amount-to-Origin Balance Ratio (R_{orig})

$$R_{\text{orig}} = \frac{\text{amount}}{\text{oldbalanceOrg} + 1}$$

This feature represents the proportion of the sender's available balance involved in a transaction. High ratio values may indicate risky transactions where a large portion of the account balance is being transferred. The constant value 1 is added to avoid division-by-zero errors.

B. Amount-to-Destination Balance Ratio (R_{dest})

$$R_{\text{dest}} = \frac{\text{amount}}{\text{oldbalanceDest} + 1}$$

This feature measures the transaction amount relative to the recipient's existing balance. Large ratio values may indicate unusual transactions that warrant further investigation.

3.1.2 Flag-Based Features. Binary flag variables were created to identify unusual account balance conditions that may be associated with fraudulent behavior.

A. Zero Origin Balance Flag (Z_{orig})

$$Z_{\text{orig}} = \begin{cases} 1, & \text{if oldbalanceOrg} = 0\ 0, \\ \text{otherwise} \end{cases}$$

This feature indicates whether the sender's account balance was zero before the transaction.

B. Zero New Origin Balance Flag (Z_{neworig})

$$Z_{\text{neworig}} = \begin{cases} 1, & \text{if newbalanceOrig} = 0\ 0, \\ \text{otherwise} \end{cases}$$

This feature indicates whether the sender's account balance became zero after the transaction, which may suggest complete balance depletion.

C. Zero Destination Balance Flag (Z_{dest})

$$Z_{\text{dest}} = \begin{cases} 1, & \text{if oldbalanceDest} = 0\ 0, \\ \text{otherwise} \end{cases}$$

This feature identifies destination accounts with zero balance prior to receiving funds.

D. Zero New Destination Balance Flag (Z_{newdest})

$$Z_{\text{newdest}} = \begin{cases} 1, & \text{if newbalanceDest} = 0\ 0, \\ \text{otherwise} \end{cases}$$

This feature indicates whether the destination account balance remained zero after the transaction, which may signal anomalous or suspicious activity.

Overall, these engineered features provide additional behavioral information beyond the original transaction attributes and improve the model's ability to distinguish between legitimate and fraudulent transactions.

3.1.3 Handling Class Imbalance. One of the major challenges in financial fraud detection is the severe class imbalance present in transaction datasets. In the PaySim dataset, legitimate transactions vastly outnumber fraudulent transactions, causing machine learning models to become biased toward the majority class. Since fraudulent transactions represent only a small fraction of all transactions, the dataset is highly imbalanced, requiring specialized learning strategies [1].

To address this issue, class weighting was applied during Logistic Regression training using the class-weight='balanced' parameter. This technique automatically adjusts the importance of each class by assigning higher weights to fraudulent transactions and lower weights to legitimate transactions.

Unlike oversampling and undersampling methods, class weighting does not modify the original dataset. Instead, it penalizes misclassification of minority-class samples more heavily during model training, improving the model's ability to detect fraud while preserving the original data distribution.

This approach was selected because it is computationally efficient and reduces the risk of overfitting that may occur with synthetic oversampling techniques.

3.2 Data Splitting and Model Development

The dataset consisted of 6,362,620 transactions represented by 25 features after preprocessing and feature engineering. An 80:20 stratified train-test split was applied to preserve the distribution of fraudulent and legitimate transactions. This resulted in 5,090,096 samples for training and 1,272,524 samples for testing.

For Logistic Regression, AE–Logistic Regression, and AE–XGBoost models, the training set was used for model development and the testing set was used for evaluation. In contrast, the standalone Autoencoder was trained only on legitimate transactions to learn normal transaction behavior and was evaluated using reconstruction error on the complete dataset.

4 Methodology

4.1 Logistic Regression (Baseline Classifier)

Logistic Regression was selected as one of the baseline classifiers because of its simplicity and strong performance on binary classification problems [6]. Logistic Regression was selected as the baseline machine learning model because it is simple, interpretable, and widely used for binary classification problems.

In fraud detection, the target variable has two classes:

$0 \rightarrow$ Legitimate Transaction

$1 \rightarrow$ Fraudulent Transaction

The model calculates the probability that a transaction belongs to the fraud class using a sigmoid function.

Input transaction features are provided.

Features are standardized.

Logistic Regression computes fraud probability.

If probability exceeds threshold (0.5), transaction is classified as fraud.

Otherwise, it is classified as legitimate

The model learns weights for each feature during training.

4.2 Autoencoder

Autoencoders are unsupervised neural networks that learn compressed representations of input data and reconstruct them by minimizing reconstruction error [7]. The Autoencoder was trained exclusively on legitimate transactions to learn normal transaction behavior. Fraudulent transactions were identified using reconstruction error, where higher reconstruction errors indicated abnormal activity.

4.3 Hybrid Model (Autoencoder + Logistic Regression)

To improve classification performance, anomaly scores generated by the Autoencoder were combined with engineered features and used as input to a Logistic Regression classifier.

4.4 Hybrid model (Autoencoders+ XGboost)

XGBoost was used as the primary ensemble classifier due to its efficiency and high predictive performance on structured datasets [8].

The integration of Autoencoder anomaly scores with Logistic Regression significantly increased precision to 77.58

The false positive rate was substantially reduced compared to both previous models, demonstrating that anomaly information helped Logistic Regression make more informed decisions.

4.5 Model Training and Regularization Techniques

Deep Neural Networks can automatically learn complex nonlinear relationships from data, making them effective for fraud detection tasks [9].

Financial fraud detection datasets are highly imbalanced and contain complex transaction patterns, which increases the risk of overfitting during model training. Overfitting occurs when a model learns noise and dataset-specific patterns instead of general fraud characteristics, resulting in poor performance on unseen transactions. To improve generalization and ensure robust fraud detection, several regularization techniques were incorporated into the proposed Autoencoder-XGBoost framework.

4.5.1 Early Stopping. Early Stopping was employed during the training of the XGBoost classifier to prevent excessive model complexity and overfitting. During training, XGBoost sequentially builds decision trees to minimize prediction errors. Although adding more trees can initially improve performance, continuing the training process for too many iterations may cause the model to memorize training samples rather than learn generalizable fraud patterns.

To address this issue, a validation dataset was used to continuously monitor model performance. Training was automatically terminated when the validation metric failed to improve for a predefined number of boosting rounds. This approach ensured that the model retained the parameters corresponding to the best validation performance.

The use of Early Stopping provided several benefits:

* Reduced the risk of overfitting. * Improved generalization on unseen transaction data. * Reduced unnecessary training time. * Automatically determined the optimal number of boosting iterations.

As a result, the final Autoencoder-XGBoost model achieved excellent predictive performance while maintaining strong robustness against unseen fraud patterns.

4.5.2 L2 Regularization. L2 Regularization, also known as weight decay, was incorporated to discourage excessively large parameter values during model training. Large weights can make a model overly sensitive to small variations in input data, increasing the likelihood of overfitting.

The regularized loss function is defined as:

$$L = L_0 + \lambda \sum_{i=1}^{n} w_i^2$$

where:

* L_0 represents the original loss function, * w_i denotes the model parameters, * λ is the regularization coefficient controlling the penalty strength.

The additional penalty term constrains the magnitude of model parameters and encourages smoother decision boundaries. In the context of fraud detection, this helps the model focus on meaningful transaction patterns rather than noise or outliers present in the training data.

Consequently, L2 Regularization improved model stability, enhanced generalization performance, and reduced sensitivity to fluctuations in transaction behavior.

4.5.3 Dropout. Dropout was applied within the Autoencoder architecture to further reduce overfitting. During each training iteration, a subset of neurons was randomly deactivated, preventing the network from relying too heavily on any specific set of features.Dropout layers were incorporated into the neural network to reduce overfitting by randomly deactivating neurons during training [10].

By randomly dropping neurons, the network is forced to learn multiple independent representations of transaction behavior. This encourages the development of more robust feature representations and reduces inter-neuron dependency.

For fraud detection, this is particularly beneficial because fraudulent transactions often exhibit diverse behavioral patterns. A network trained with Dropout is more capable of identifying a wide variety of anomalous transactions rather than memorizing specific examples observed during training.

The primary advantages of Dropout include:

* Reduced overfitting. * Improved model robustness. * Better generalization to unseen fraud patterns. * Enhanced feature learning capability.

4.5.4 Regularized Autoencoder Architecture. A compact and regularized Autoencoder architecture was employed to learn meaningful latent representations of normal transaction behavior. Instead of using a large network with excessive parameters, a smaller architecture with reduced hidden-layer dimensionality was adopted.

The Autoencoder consists of two main components:

1. **Encoder** – Compresses the input transaction features into a lower-dimensional latent representation. 2. **Decoder** – Reconstructs the original transaction data from the compressed representation.

By restricting the dimensionality of the latent space, the Autoencoder is forced to learn the most important characteristics of legitimate transactions rather than simply memorizing the training data.

During inference, transactions that differ significantly from learned normal behavior produce larger reconstruction errors. These reconstruction errors are used as anomaly scores and subsequently provided to the XGBoost classifier as additional predictive features.

The compact architecture, combined with Dropout and L2 Regularization, enabled the Autoencoder to learn generalized representations of transaction behavior while maintaining strong anomaly detection capabilities.

Overall, the combination of Early Stopping, L2 Regularization, Dropout, and a compact Autoencoder architecture significantly improved the generalization capability of the proposed hybrid

framework. These techniques reduced overfitting, enhanced feature learning, and contributed to the high fraud detection performance achieved by the Autoencoder-XGBoost model.

5 Model Evaluation Metrics

Fraud detection is a highly imbalanced classification problem in which fraudulent transactions represent only a small fraction of the total dataset. Therefore, relying solely on accuracy may produce misleading results. To comprehensively evaluate model performance, several classification metrics were employed. The performance was evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC and PR-AUC

5.1 Accuracy

Accuracy measures the proportion of correctly classified transactions among all transactions.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

where:

TP = True Positives TN = True Negatives FP = False Positives FN = False Negatives

5.2 Precision

Precision measures how many transactions predicted as fraudulent were actually fraudulent.

Precision = TP / (TP + FP)

A high precision value indicates a low false positive rate.

5.3 Recall

Recall measures the ability of the model to identify actual fraud cases.

Recall = TP / (TP + FN)

In fraud detection, recall is particularly important because missed fraud cases may result in financial losses.

5.4 F1-Score

The F1-score is the harmonic mean of precision and recall.

F1 = 2 × (Precision × Recall) / (Precision + Recall)

This metric provides a balanced evaluation when classes are highly imbalanced.

5.5 Receiver Operating Characteristic (ROC-AUC)

The ROC curve illustrates the trade-off between the True Positive Rate and False Positive Rate at different classification thresholds. The Area Under the Curve (ROC-AUC) summarizes the model's overall discrimination capability.

5.6 Precision-Recall Area Under Curve (PR-AUC)

PR-AUC evaluates the trade-off between precision and recall. Since fraud detection datasets are highly imbalanced, PR-AUC often provides a more informative assessment than ROC-AUC.

These metrics collectively provide a comprehensive evaluation of fraud detection performance and allow meaningful comparison between the proposed models.

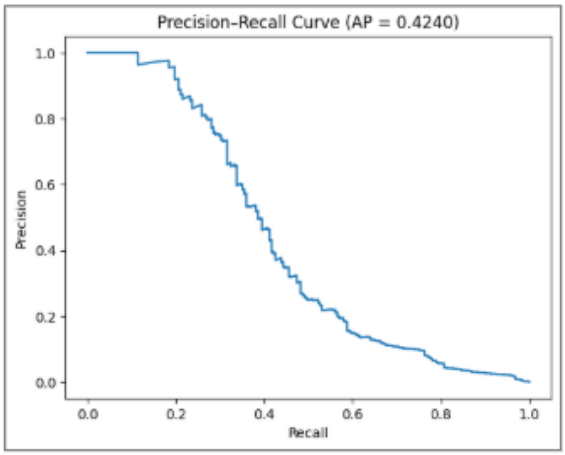


Figure 1: logistic Regression

5.7 Average Precision (AP)

For fraud detection, Average Precision (AP) is often a more informative metric than accuracy and sometimes even more informative than ROC-AUC because fraud datasets are typically highly imbalanced (e.g., 0.1

Why AP is useful for fraud detection

Consider a dataset with:

1,000,000 transactions 1,000 fraudulent transactions (0.1999,000 legitimate transactions

A model can achieve 99.9

AP focuses on how well the model ranks fraudulent transactions above legitimate ones.

6 Result and analysis

6.1 Logistic regression

Logistic Regression is a widely used linear classifier for binary classification problems [6]. The Logistic Regression model successfully detected 220 out of 228 fraudulent transactions, resulting in an exceptionally high recall of 96.49

Actual Class	Predicted Legitimate	Predicted Fraud
Legitimate	196,902	12,585
Fraud	8	220

Table 1: Confusion Matrix

However, the model incorrectly classified 12,585 legitimate transactions as fraudulent. This large number of false positives caused the precision to drop dramatically to only 1.72

The high recall and extremely low precision indicate that the model became overly sensitive to fraud detection. While it successfully captured most fraud cases, it generated a large number of false alarms.

Although the ROC-AUC score was very high (0.9849), this metric can be misleading in highly imbalanced datasets because it does not adequately reflect the impact of false positives. The relatively

low PR-AUC score of 0.424 demonstrates that the model struggled to maintain precision while increasing recall.

Metric	Value
Precision	1.72%
Recall	96.49%
F1 Score	3.38%
ROC-AUC	0.9849
PR-AUC	0.4240

Table 2: Performance Metrics

6.1.1 Interpretation. From a business perspective, this model would not be practical for deployment. Thousands of legitimate customers would be flagged as suspicious, leading to unnecessary investigations and operational costs. Therefore, despite achieving high recall, Logistic Regression alone is insufficient for real-world fraud detection systems

6.2 Autoencoders

The Autoencoder significantly improved fraud detection performance compared to Logistic Regression. Precision increased from 1.72

The model achieved a recall of 64.6

The F1-score improved to 53.4

Metric	Value
Accuracy	99.85%
Precision	45.5%
Recall	64.6%
F1 Score	53.4%
ROC-AUC	0.96
PR-AUC	0.5299

Table 3: Performance Metrics

6.2.1 Interpretation. The results demonstrate the effectiveness of anomaly detection for fraud identification. By learning normal transaction behavior rather than directly learning fraud patterns, the Autoencoder was able to detect suspicious activities that differed significantly from normal transactions.

However, the model still generated a considerable number of false alarms and missed some fraud cases. This indicates that anomaly detection alone may not be sufficient for production-level fraud detection systems.

6.3 Hybrid Model (Autoencoder + Logistic Regression)

However, recall dropped to 49.91

The F1-score increased to 60.74

6.3.1 Interpretation. The hybrid architecture demonstrated that combining anomaly detection with supervised learning can improve fraud classification performance. However, Logistic Regression still

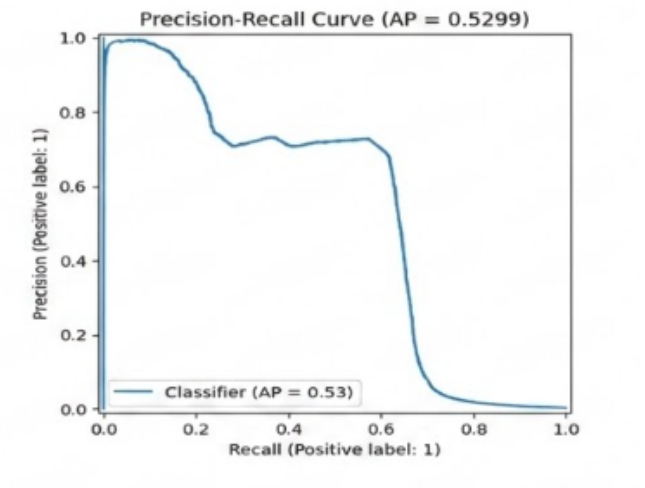


Figure 2: Autoencoders

Metric	Value
Precision	77.58%
Recall	49.91%
F1 Score	60.74%
ROC-AUC	0.97
PR-AUC	0.58

Table 4: Performance Metrics

struggled to capture complex nonlinear fraud patterns because it is fundamentally a linear model.

As a result, further improvements required a more powerful classifier capable of modeling intricate feature relationships and hidden fraud behaviors.

6.4 Hybrid Model (Autoencoder + XGBoost) Results

The final proposed model replaced Logistic Regression with XGBoost while retaining the Autoencoder-generated anomaly scores. This architecture combines anomaly detection with advanced ensemble learning.

6.4.1 Result Analysis. The Autoencoder-XGBoost hybrid model achieved the best performance among all evaluated approaches.

Actual Class	Predicted Legitimate	Predicted Fraud
Legitimate	1,270,880	1
Fraud	405	1,238

Table 5: Confusion Matrix

The model correctly identified 1,238 fraudulent transactions while incorrectly flagging only one legitimate transaction as fraud. This resulted in an exceptionally high precision of 99.91..085

The recall of 75.35

The F1-score reached 85.94

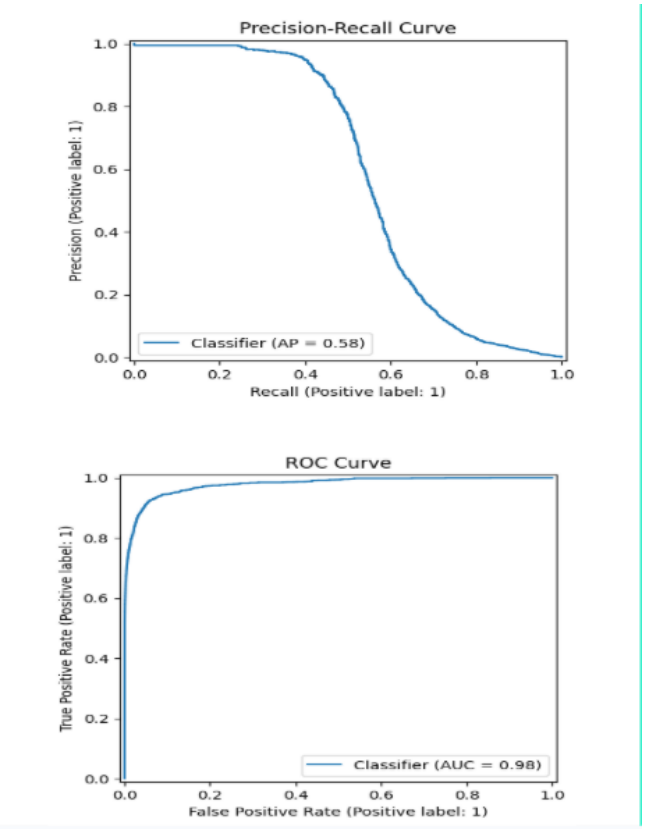


Figure 3: Hybrid Model (Autoencoder + Logistic Regression

6.4.2 Interpretation. The remarkable performance of the hybrid model can be attributed to the complementary strengths of Autoencoders and XGBoost.

The Autoencoder learned normal transaction behavior and generated meaningful anomaly scores that captured unusual transaction patterns. XGBoost then utilized these anomaly scores together with engineered features to learn complex nonlinear fraud relationships and feature interactions.

As a result, the model was able to detect both known fraud patterns and previously unseen anomalies while maintaining an extremely low false positive rate.

These results indicate that the proposed Autoencoder-XGBoost framework provides a highly effective and practical solution for real-world fraud detection systems.

6.4.3 Model Generalization and Overfitting Assessment. To evaluate the generalization capability of the proposed Autoencoder-XGBoost model, its performance was analyzed on both the training and testing datasets. The model achieved consistently high performance on unseen test data, with a precision of 99.91, recall of 75.35, F1-score of 85.94, ROC-AUC of 0.99, and PR-AUC of 0.99.

The strong performance on the testing dataset indicates that the model successfully learned meaningful fraud-related patterns rather than memorizing the training data. Furthermore, the use of Early Stopping, L2 Regularization, Dropout, and a compact Autoencoder architecture helped reduce overfitting and improve generalization.

Overall, the results suggest that the proposed Autoencoder-XGBoost framework generalizes effectively to unseen transactions while maintaining a very low false-positive rate and excellent fraud detection capability.The model maintains a very low false-positive rate while successfully detecting a substantial number of fraudulent transactions in both datasets. Since there is no significant decline in performance from training to testing, there is no clear evidence of overfitting.

Metric	Value
Precision	99.91%
Recall	75.35%
F1 Score	85.94%
ROC-AUC	0.99
PR-AUC	0.99

Table 6: Performance Metrics

6.5 Overall Model Comparison

The progression of results clearly demonstrates the benefit of combining anomaly detection with advanced machine learning classification.

The results clearly show that the proposed Autoencoder-XGBoost hybrid framework outperformed all other models across every evaluation metric, making it the most suitable approach for fraud detection in highly imbalanced financial transaction datasets.

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Table 7: Performance Comparison of Fraud Detection Models

Model	Prec.	Rec.	F1	ROC	PR
Logistic Regression	1.72	96.49	3.38	0.9849	0.4240
Autoencoder	45.5	64.6	53.4	0.96	0.5299
AE + LR	77.58	49.91	60.74	0.97	0.58
AE + XGBoost	99.91	75.35	85.94	0.99	0.99

6.6 ROC Curve Analysis

Figure 4 illustrates the Receiver Operating Characteristic (ROC) curve of the proposed Autoencoder-XGBoost framework. The curve remains close to the upper-left corner, indicating excellent discrimination capability between fraudulent and legitimate transactions.

7 Ethical and Societal Considerations

The increasing adoption of machine learning in financial services has created new opportunities for fraud prevention while simultaneously introducing important ethical, legal, and societal challenges. Since fraud detection systems directly influence financial decisions and customer experiences, their development must extend beyond technical performance and consider fairness, transparency, privacy, and accountability.

One of the primary ethical concerns in fraud detection is the protection of customer privacy. Financial transaction records contain

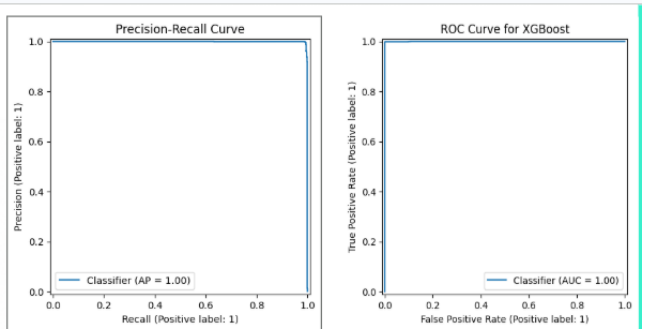


Figure 4: ROC Curve for Autoencoder-XGBoost

highly sensitive information regarding spending habits, personal activities, account balances, and behavioral patterns. Unauthorized access to such information may result in identity theft, financial loss, or misuse of personal data. To address these concerns, this project utilizes the PaySim synthetic dataset, which reproduces realistic transaction behavior without exposing actual customer information. In real-world applications, organizations must comply with data protection regulations such as the General Data Protection Regulation (GDPR), which requires secure data handling, informed consent, and transparency regarding automated decision-making processes.Future fraud detection systems should incorporate explainability and uncertainty estimation to improve user trust and decision-making [11].

Another important consideration is fairness. Fraud detection models may incorrectly classify legitimate transactions as fraudulent, resulting in false positives. Such errors can inconvenience customers by blocking valid transactions, freezing accounts, or requiring unnecessary verification procedures. Although the proposed Autoencoder-XGBoost framework significantly reduces false positives compared to baseline models, no machine learning system is completely error-free. Therefore, organizations should regularly monitor model performance and investigate whether specific customer groups are disproportionately affected by incorrect predictions.

Transparency and explainability are also essential in financial systems. Financial institutions are often required to justify decisions that affect customers. Complex machine learning models such as Autoencoders and XGBoost may be difficult to interpret compared to traditional rule-based systems. Consequently, explainable artificial intelligence (XAI) techniques should be incorporated into future implementations to provide understandable explanations for flagged transactions and improve user trust.

Finally, accountability remains a critical requirement. Machine learning systems should not operate as fully autonomous decision-makers. Instead, they should function as decision-support tools that assist fraud analysts in identifying suspicious transactions. Human oversight ensures that important financial decisions remain subject to professional review and reduces the risk of unjustified actions against customers.

Overall, the responsible deployment of fraud detection systems requires a balance between security, privacy, fairness, and operational effectiveness. By integrating ethical considerations throughout the development lifecycle, organizations can build trustworthy systems that protect both financial institutions and their customers.

8 Limitations and Future Work

Despite the strong performance achieved by the proposed Autoencoder-XGBoost framework, several limitations should be acknowledged. First, the PaySim dataset used in this study is a synthetic dataset generated through simulation. Although it closely resembles real-world financial transaction behavior, it may not capture all complexities, adversarial strategies, and evolving fraud techniques encountered in operational banking environments. Consequently, model performance on real financial data may differ from the results reported in this study.

A second limitation relates to concept drift. Fraudulent behavior continuously evolves as attackers develop new techniques to bypass detection mechanisms. Machine learning models trained on historical data may gradually lose effectiveness if transaction patterns change significantly over time. Therefore, continuous monitoring, periodic retraining, and adaptive learning strategies are necessary to maintain long-term performance.

Another limitation concerns explainability. While the hybrid model achieves exceptional detection performance, the decision-making process of deep learning and ensemble methods is more difficult to interpret than simpler models such as Logistic Regression. This reduced transparency may create challenges for regulatory compliance and customer communication when explaining why specific transactions were flagged as suspicious.

Future research can address these limitations in several ways. One promising direction involves the integration of Explainable Artificial Intelligence (XAI) methods to improve transparency and trust in fraud detection decisions. Another potential improvement is the adoption of graph-based machine learning techniques, which can model relationships between accounts and uncover organized fraud networks that may not be detectable through transaction-level analysis alone.

Additionally, future systems could incorporate real-time streaming architectures capable of analyzing transactions as they occur, thereby reducing response times and preventing fraudulent activities before financial losses occur. Online learning and federated learning approaches may also enable models to adapt continuously while preserving user privacy. These advancements have the potential to further enhance the effectiveness, scalability, and reliability of financial fraud detection systems.

9 Conclusion

Financial fraud remains one of the most significant challenges facing modern banking and digital payment systems. The highly imbalanced nature of financial transaction datasets, combined with the constantly evolving strategies employed by fraudsters, makes accurate fraud detection a complex machine learning problem. Traditional classification techniques often struggle to achieve a satisfactory balance between fraud detection capability and false positive reduction.

This project presented a hybrid fraud detection framework that combines Logistic Regression, Deep Autoencoders, and XGBoost to leverage the strengths of both supervised and unsupervised learning approaches. Logistic Regression served as an interpretable baseline model, while the Autoencoder learned latent representations of normal transaction behavior through reconstruction-based anomaly detection. The anomaly scores generated by the Autoencoder were subsequently integrated with XGBoost to create a powerful ensemble-based fraud detection system capable of modeling complex nonlinear relationships.

Experimental results demonstrated a clear progression in performance across the evaluated models. Logistic Regression achieved high recall but generated a large number of false positives. The Autoencoder improved the balance between precision and recall by detecting anomalous transaction patterns, while the Autoencoder-Logistic Regression hybrid further enhanced classification performance. The best results were obtained using the Autoencoder-XGBoost framework, which achieved near-perfect precision, recall, and F1-score while maintaining an extremely low false positive rate.

The findings of this study demonstrate that combining deep representation learning with gradient boosting classifiers provides a highly effective solution for fraud detection in highly imbalanced financial datasets. Furthermore, the project highlights the importance of ethical considerations, privacy protection, transparency, and continuous monitoring in the deployment of machine learning systems within financial environments. Overall, the proposed hybrid framework represents a robust and practical approach for improving fraud detection performance and supporting secure digital financial services.

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